

Banglalink FWA Network Measurement SDK - Integration Guide

This document provides a comprehensive guide for integrating the FWA Network Measurement SDK into host applications. It covers setup, initialization, manifest configurations, and common troubleshooting steps.

Project Configuration

1.1 SDK Dependency

Ensure the SDK is added to your project dependencies.

```
// build.gradle.kts (Module)
dependencies {

    implementation("com.github.primetechsolutions:FWA_Network_Measuremen
t_SDK:X.X.X") // Replace with SDK Version
}
```

```
//settings.gradle.kts (Project Level)
maven {
    url = uri("https://jitpack.io")
}
```

1.2 Manifest Configuration & WorkManager

If this block exists from a previous (e.g., network measurement) SDK and is not required anymore, it can be safely removed.

If your app does not include this configuration, no action is required—WorkManager will initialize automatically.

If your host app removes the default WorkManager initializer using `tools:node="remove"`, automatic initialization will be disabled. In that case, you **must manually initialize WorkManager** in your `Application` class (app starter).

```
<!-- AndroidManifest.xml -->
<provider
    android:name="androidx.startup.InitializationProvider"
    android:authorities="${applicationId}.androidx-startup"
    tools:node="remove" />
```

2. Initialization

The SDK must be initialized before triggering any network captures. Because the SDK manages permissions, it must be bound to the lifecycle of an Activity or Fragment.

Initialization Rules:

You MUST call `NetworkDataUploader().init(...)` inside `onCreate()` of your Activity or Fragment.

```
private lateinit var FWANetworkDataUploader:
FWANetworkDataMeasurement()

override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
    // Initialize SDK for activity/fragment
    FWANetworkDataUploader = FWANetworkDataMeasurement()
    FWANetworkDataUploader.init(this, "AppName")
}
```

3. Capturing Network Data

Parameter	Type	Requirement	Description
msisdn	String	Required	User mobile number.
integratedAppVersion	String	Required	Host app version.
sdkInitiateTimeStamp	String	Required	Format: yyyy-MM-dd'T'HH:mm:ss.
integratedAppEventName	String	Required	Unique name for the event.
userLatitude	Double	Optional	Fallback latitude.

Parameter	Type	Requirement	Description
userLongitude	Double	<i>Optional</i>	Fallback longitude.

startUploading() Parameters

FWA Diagnostic Measurement

```
FWANetworkDataUploader.startFWAMeasurement(
    msisdn = "01XXXXXXXXX",
    integratedAppVersion = "1.0.0",
    sdkInitiateTimeStamp = "2024-02-27T22:15:50",
    integratedAppEventName = "Manual_Diagnostic",
) { success, status ->
    // status.message contains the JSON response
    Log.d("SDK", "FWA Result: ${status.response}")
}
```

4. Understanding Responses : FTP Response Examples

Error Response (e.g., SDK not initialized)

```
{
  "status": "Failed",
  "statusCode": 400,
  "message": "SDK not initialized"
}
```

Success/Pass Response

The status.message returns a JSON string containing the diagnostic values:

```
{
  "status": "Success",
  "testResult": "Pass",
  "statusCode": 200,
  "message": "Your network assessment successful.",
  "data": {
    "assessmentId": 119,
    "networkData": { "RSRP": -105, "SNR": 18, "RSRQ": -3 },
    "cellInfo": {
      "cellName": "DHK_K1469_1",
      "eNodeBName": "DHK_K1469",
      "nbhDlThroughputMbps": 11.5,
      "nbhTrafficGB": 0.5
    },
    "speedPair": { "ulSpeedKbps": 5000.00, "dlSpeedKbps": 5000.00 },
    "userInfo": {
      "msisdn": "01XXXXXXXXX",
      "deviceManufacture": "OnePlus",
      "deviceModel": "ONEPLUSA6000",
      "deviceOsVersion": "30",
      "latitude": 23.7515,
      "longitude": 90.3933
    }
  }
}
```

Success/Failed Response

The status.message returns a JSON string containing the diagnostic values:

```
{
  "status": "Success",
  "testResult": "Failed",
  "statusCode": 200,
  "message": "Your network assessment failed.",
  "data": {
    "assessmentId": 120,
    "networkData": { "RSRP": -120, "SNR": 18, "RSRQ": -3 },
    "cellInfo": {
      "cellName": "DHK_K1469_1",
      "eNodeBName": "DHK_K1469",
      "nbhDlThroughputMbps": 11.5,
      "nbhTrafficGB": 0.5
    },
    "speedPair": { "ulSpeedKbps": 6763.00, "dlSpeedKbps": 3487.00 },
    "userInfo": {
      "msisdn": "01XXXXXXXXX",
      "deviceManufacture": "OnePlus",
      "deviceModel": "ONEPLUSA6000",
      "deviceOsVersion": "30",
      "latitude": 23.7515,
      "longitude": 90.3933
    }
  }
}
```

Failed Response (Example: Permissions Missing)

```
{
  "status": "Failed",
  "testResult": "Failed",
  "statusCode": 400,
  "message": "Permission are required for FTP Capture",
  "data": null
}
```

Failed Response (Example: Timeout)

```
{
  "status": "Failed",
  "testResult": "Failed",
  "statusCode": 408,
  "message": "Network assessment timed out. Please check your internet connection.",
  "data": null
}
```

Failed Response (Example: Non Banglalink 4G)

```
{
  "status": "Failed",
  "testResult": "Failed",
  "statusCode": 400,
  "message": "FTP Capture is only supported on Banglalink 4G (LTE) technology."
}
```

Failed Response (Example: Non Mobile Network)

```
{
  "status": "Failed",
  "testResult": "Failed",
  "statusCode": 400,
  "message": "FTP Capture requires mobile data. Please disable Wi-Fi and ensure Banglalink 4G is active."
}
```

5. Requirements

Required Permissions

- **GPS_ENABLE, INTERNET, ACCESS_FINE_LOCATION, ACCESS_COARSE_LOCATION, READ_PHONE_STATE.**

[!IMPORTANT] Without these permissions, the app may not be able to collect accurate data or function correctly.

Platform Support

This SDK is exclusively supported on the **Android Platform**.

6. Best Practices

[!WARNING] **Production Readiness:** Perform cross-testing with the SDK before releasing any app update to production to verify end-to-end functionality and data consistency.

About the Network Measurement SDK The Network Measurement SDK (Android) is a comprehensive tool for real-time mobile network analysis. It provides integrated signal telemetry, throughput measurement, and diagnostic reporting to help host applications maintain optimal connectivity for their users.

Support & Integration Sync

If needed, for better understanding regarding the SDK integration, please feel free to arrange a meeting with the SDK Development Team. We'll be happy to assist and walk you through the implementation.